Towards In-the-wild 3D Plane Reconstruction from a Single Image

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Abstract

*Reconstructing 3D planes from a single image is fundamental to scene understanding, yet existing methods are typically trained and evaluated on a single domain, limiting their generalization to unseen environments. We present the **ZeroPlane Planar Benchmark**, a large-scale mixed-domain benchmark spanning 14 datasets with approximately 560,000 high-resolution (640×480) dense plane annotations covering diverse indoor and outdoor scenes. Building on this benchmark, we propose **ZeroPlane**, a Transformer-based method that pairs a DINOv2 encoder with a DPT-style pixel decoder and introduces a decoupled normal/offset prediction strategy via exemplar-guided classification-then-regression, together with geometry-enhanced plane embeddings. Trained on our mixed-domain corpus, ZeroPlane achieves strong zero-shot generalization: it consistently outperforms the PlaneRecTR baseline on plane recall across four held-out evaluation sets—NYUv2, 7-Scenes, ApolloScape, and ParallelDomain—without any domain-specific fine-tuning. Our benchmark and method establish a new foundation for in-the-wild 3D plane reconstruction research.*

1. Introduction

Three-dimensional plane reconstruction from a single RGB image is a core building block for numerous downstream tasks, including augmented reality, robot navigation, and architectural scene modeling. Planar surfaces dominate man-made environments—walls, floors, tables, and roads—and recovering their geometry provides a compact yet informative scene representation. Despite steady progress, current single-image plane reconstruction methods [10, 16, 20, 27] are overwhelmingly trained and evaluated on ScanNet [5] alone, raising concerns about how well they transfer to new domains.

This single-domain paradigm carries two fundamental limitations. First, the resulting models are brittle: a detector trained exclusively on ScanNet indoor scenes struggles with outdoor road scenes and vice versa. Second, evaluation on a single benchmark inflates confidence in generalization ability because the test distribution closely matches the training distribution. In the broader field of geometric estimation, mixed-dataset training has proven critical for zero-shot transfer in monocular depth [13] and surface normal prediction, yet no analogous large-scale effort exists for plane reconstruction.

To close this gap, we introduce the **ZeroPlane Planar Benchmark**, a unified benchmark that aggregates and harmonizes plane annotations from 14 heterogeneous sources. The benchmark comprises roughly 560,000 images with dense, per-pixel plane labels at a consistent 640×480 resolution. Training sources span photorealistic indoor simulators (Replica [19], HM3D [25]), real-world indoor scans (ScanNetv1/v2 [5], Matterport3D [3], DIODE [21], Taskonomy [28]), and synthetic outdoor environments (Synthia [15], Virtual KITTI [1, 6], Sanpo [22]). Four additional datasets—NYUv2 [18], 7-Scenes [17], ApolloScape [8], and ParallelDomain [7, 12]—are reserved exclusively for zero-shot evaluation.

Leveraging this benchmark, we propose **ZeroPlane**, a Transformer-based architecture designed for cross-domain plane reconstruction. ZeroPlane employs a frozen DINOv2 [11] encoder paired with a DPT-style pixel decoder and introduces two key innovations: (i) a decoupled normal/offset prediction pipeline that uses exemplar-guided classification followed by regression to handle the wide range of plane parameters across domains, and (ii) geometry-enhanced plane embeddings that inject structural priors into the query representation. Trained with domain-balanced sampling on our mixed corpus, ZeroPlane consistently improves zero-shot plane recall on all four held-out evaluation sets relative to the PlaneRecTR [16] baseline.

In summary, our contributions are as follows:

- We construct the **ZeroPlane Planar Benchmark**, the first large-scale, mixed-domain benchmark for single-image 3D plane reconstruction, spanning 14 datasets with $\sim$560k dense plane annotations.
- We propose **ZeroPlane**, a Transformer-based method featuring exemplar-guided decoupled plane parameter prediction and geometry-enhanced plane embeddings.
- We demonstrate that mixed-domain training yields consistent zero-shot improvements on four unseen evaluation datasets covering both indoor and outdoor scenes.

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2. Related Work

2.1. Single-Image 3D Plane Reconstruction

Early learning-based approaches formulated plane detection as either a per-pixel prediction task or an instance-level detection problem. PlaneNet [9] proposed an end-to-end CNN that directly regresses plane parameters and segmentation masks. PlaneRCNN [10] adopted a two-stage detect-then-refine paradigm built on Mask R-CNN, improving mask quality through a warping-based refinement module. PlaneAE [27] introduced an associative-embedding approach to group pixels into plane instances in a bottom-up manner. More recently, Transformer-based architectures have been applied: PlaneTR [20] cast plane reconstruction as a set prediction problem using DETR-style queries, and PlaneRecTR [16] further improved upon this with a unified query-based framework. PlanarRecon [24] extended the task to incremental 3D reconstruction from video. Despite these architectural advances, nearly all methods are trained and benchmarked on ScanNet [5], leaving cross-domain generalization largely unexplored.

Beyond detection accuracy, plane representations have proven particularly useful for a wide range of downstream 3D scene-understanding tasks. Planar constraints have been used to regularize monocular depth completion [10] and to accelerate multi-view stereo by replacing dense voxel grids with compact planar primitives. In augmented reality and robotics, real-time plane tracking underpins surface interaction, occlusion handling, and semantic mapping of indoor workspaces. Autonomous driving systems likewise rely on road-plane estimation for ground-height normalization and drivable-area segmentation. Despite this broad applicability, the single-domain training paradigm prevalent in current methods severely limits deployability across environments: a model trained exclusively on indoor apartment scans is unlikely to recover road planes in driving scenes without domain-specific fine-tuning. Our work directly addresses this limitation by targeting zero-shot cross-domain generalization as a primary design objective, rather than treating it as an afterthought.

2.2. Mixed-Dataset Training for Geometric Estimation

Training on diverse data mixtures has been transformative for monocular depth estimation. Ranftl *et al*. [13] demonstrated that mixing multiple depth datasets with careful scale-and-shift alignment enables zero-shot transfer to unseen domains. Similar strategies have since been adopted for surface normal estimation and multi-task geometric prediction. Foundation vision encoders such as DINOv2 [11], pre-trained on large uncurated image collections, provide robust features that further facilitate cross-domain transfer.

Building on these foundations, recent work has further scaled mixed-dataset training for geometric estimation. UniDepth [**?**] introduced a camera-model-aware architecture enabling metric depth prediction without calibration, while Depth Anything [26] exploited large-scale unlabeled imagery via pseudo-label distillation to improve zero-shot depth accuracy. For surface normals, DSINE [**?**] incorporated pixel-wise ray directions as geometric priors, yielding more consistent cross-domain predictions. These advances share a common theme: distributional breadth during training is essential for robust generalization, a principle we extend to plane instance reconstruction. Our work brings this mixed-training philosophy to the plane reconstruction task for the first time.

2.3. Transformer-Based Detection and Segmentation

The success of DETR and its variants has motivated query-based architectures across detection and segmentation tasks. Mask2Former [4] unified panoptic, instance, and semantic segmentation under a single masked-attention Transformer framework. In our pipeline, Mask2Former serves a dual role: it provides panoptic pseudo-labels for datasets lacking semantic annotations, and its architectural principles inspire our query-based plane decoder.

2.4. Benchmarks for Geometric Scene Understanding

Large-scale benchmarks have been instrumental in driving progress in depth estimation, surface normal prediction, and 3D reconstruction. However, existing plane reconstruction benchmarks are limited to single datasets—primarily ScanNet—with a narrow domain focus. Our ZeroPlane Planar Benchmark addresses this by unifying 14 datasets under a consistent annotation protocol, enabling both large-scale training and rigorous zero-shot evaluation. Prior multi-task benchmarks such as Taskonomy [28] have demonstrated that joint training across geometric tasks can improve individual task performance, yet plane instance detection—which requires simultaneously accurate segmentation boundaries and precise 3D parameter estimation—has remained underserved. Our benchmark fills this gap by providing a standardized evaluation suite that measures both segmentation quality and geometric fidelity across domains.

3. Datasets

We introduce the **ZeroPlane Planar Benchmark**, a large-scale, mixed-domain benchmark for single-image 3D plane reconstruction. The benchmark aggregates 14 datasets spanning real and synthetic, indoor and outdoor environments, totaling approximately 560,000 images with dense plane annotations at a unified resolution of 640×480. Tab. 1 provides a detailed breakdown of each data source.

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Table 1. Statistics of the datasets used in our work. Top: Datasets used for training and validation. Bottom: Datasets used for zero-shot evaluation.

Dataset	Indoor	Outdoor	Label	#Training Images	#Validation Images	#Planes
ScanNetv1 [5]	✓		RGB-D	47923	719	281268
ScanNetv2 [5]	✓		RGB-D	140773	859	1445183
Matterport3D [3]	✓		RGB-D	20854	943	227157
Replica [19]	✓		Synthetic	5533	400	90466
HM3D [25]	✓		Synthetic	79994	2000	1298635
DIODE (Indoor Split) [21]	✓		RGB-D	7350	731	37447
Taskonomy [28]	✓		RGB-D	123740	499	1450599
Synthia [15]		✓	Synthetic	24636	999	143698
Virtual KITTI [1, 6]		✓	Synthetic	17720	970	71171
Sanpo (Synthetic Split) [22]		✓	Synthetic	89854	488	647530
NYUv2 [18]	✓		RGB-D	-	654	4849
7-Scenes [17]	✓		RGB-D	-	758	5433
ApolloScape [8] (Stereo Split)		✓	Stereo	-	999	4241
ParallelDomain [7, 12]		✓	Synthetic	-	356	2350

3.1. Data Sources

Our training corpus draws from ten datasets covering diverse scene types. Indoor sources include ScanNetv1/v2 [5], Matterport3D [3], Replica [19], HM3D [25], the indoor split of DIODE [21], and Taskonomy [28]. Outdoor and synthetic sources include Synthia [15], Virtual KITTI [1, 6], and the synthetic split of Sanpo [22]. Four datasets are held out exclusively for zero-shot evaluation: NYUv2 [18] and 7-Scenes [17] for indoor scenes, and ApolloScape [8] and ParallelDomain [7, 12] for outdoor scenes.

3.2. Plane Annotation Pipeline

For each image, we construct dense plane annotations through a three-stage pipeline. First, we back-project the depth map into a 3D point cloud using the provided camera intrinsics. Second, we obtain per-image panoptic segmentation masks: datasets with ground-truth semantic labels use those directly, while for datasets lacking such annotations we generate panoptic pseudo-labels using a pretrained Mask2Former [4] model. Third, within each panoptic segment we fit a 3D plane to the back-projected points using RANSAC, retaining segments whose inlier ratio exceeds a quality threshold. All annotations are stored as dense per-pixel plane masks at 640×480 resolution, and camera intrinsics are consistently handled across datasets to ensure compatible coordinate frames.

3.3. Evaluation Protocol

We evaluate plane reconstruction along two complementary axes. **Segmentation quality** is measured via Rand Index (RI), Variation of Information (VOI), and Segmentation Covering (SC). **Plane geometry accuracy** is assessed through plane recall at multiple thresholds: depth error thresholds of 0.05 m, 0.1 m, and 0.6 m for indoor scenes and 1 m, 3 m, and 10 m for outdoor scenes, together with normal angle thresholds of $5°$, $10°$, and $30°$. During training, we apply domain-balanced sampling to prevent large datasets from dominating gradient updates, ensuring that the model receives approximately equal exposure to each source domain.

4. Methodology

We present ZeroPlane, a unified framework for single-image 3D plane reconstruction that generalizes across diverse indoor and outdoor scenes. As illustrated in Fig. 1, our architecture builds upon a transformer-based detection paradigm [2, 4] and introduces three key design choices: (1) a DPT-style pixel decoder for multi-scale feature extraction, (2) decoupled geometry prediction via exemplar-guided classification-then-regression for plane normals and offsets, and (3) geometry-enhanced embeddings that inject pixel-level depth and normal cues into the plane query decoding process.

4.1. Backbone and Pixel Decoder

We adopt DINOv2 [11] as our image encoder, leveraging its strong visual representations pretrained on large-scale data. We experiment with both ViT-Base and ViT-Large variants. Additionally, we explore DUSt3R [23] as an alternative backbone that provides geometry-aware features. The encoder outputs multi-scale feature maps that are fed into a DPT-style pixel decoder [14], which progressively fuses features from different transformer layers through upsampling and concatenation to produce high-resolution, multi-scale pixel features. These features serve as the key and value inputs for the subsequent transformer decoder.

4.2. Plane Query Decoding

Following Mask2Former [4], we employ a transformer decoder that takes a set of learnable plane queries and refines them through masked cross-attention with the multi-scale pixel features. Each decoded plane query encodes both the identity and geometric properties of a predicted plane instance. The queries are matched to ground-truth planes during training via bipartite matching [2], and each matched query produces a plane segmentation mask through dot-product attention with the pixel features, along with the plane's geometric parameters.

4.3. Decoupled Geometry Prediction

Directly regressing plane parameters (normal vector $\mathbf{n}$ and offset d, where the plane equation is $\mathbf{n}^\top \mathbf{x} = d$) proves challenging due to the wide range of possible values across indoor and outdoor scenes. We decouple the prediction into separate normal and offset heads, each employing an exemplar-guided classification-then-regression strategy.

For the **normal head**, we pre-compute $K_n = 7$ exemplar directions via K-means clustering on training data normals. The head first classifies each plane query into one of the K_n exemplars, then regresses a residual vector to refine the prediction. Similarly, the **offset head** uses $K_d = 20$ exemplar offset values, with a split point at 20 m to accommodate the distinct depth distributions of indoor and outdoor environments. The final plane parameter is obtained as $\boldsymbol{\pi} = \mathbf{n}/d$.

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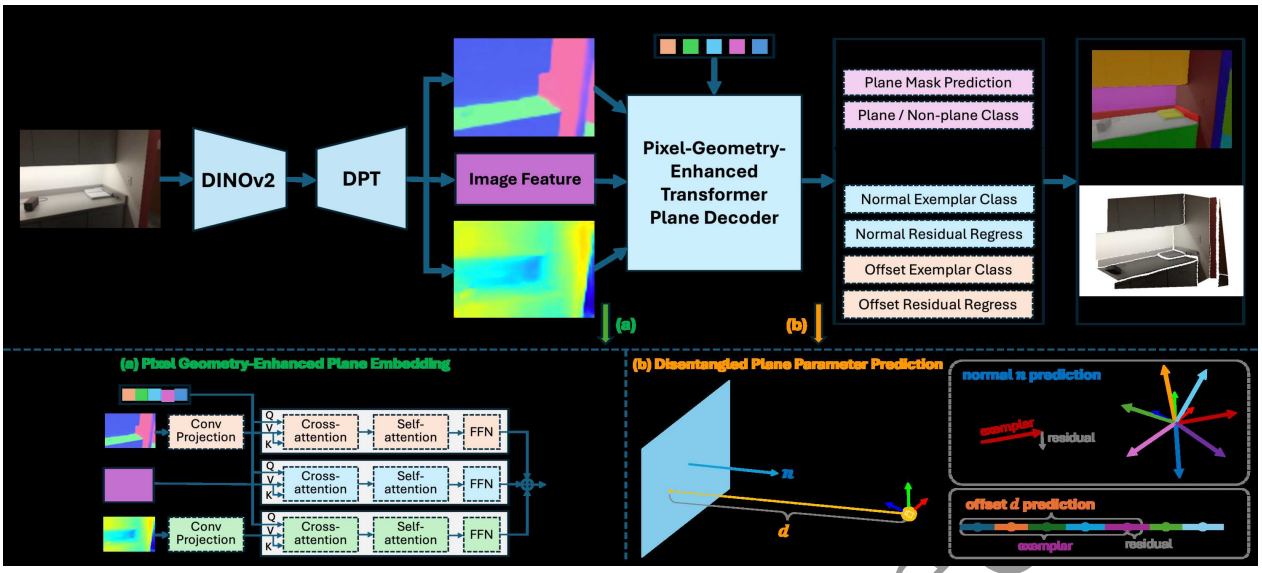

Figure 1. Overview of the proposed **ZeroPlane** framework. A DINOv2 (or DUSt3R) backbone extracts multi-scale features, refined by a DPT-style pixel decoder. A Transformer decoder with geometry-enhanced queries predicts plane masks and decoupled normal/offset parameters via exemplar-guided classification-then-regression heads. Auxiliary pixel-level depth and surface normal heads provide geometry embeddings fused into plane queries via cross-attention.

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This classification-then-regression scheme constrains the output space, reducing the difficulty of cross-domain generalization while maintaining precise predictions through residual refinement.

The exemplar codebooks are constructed offline by applying K-means clustering to all training-set plane normals and offsets prior to model initialization. The seven normal exemplars capture the dominant axis-aligned orientations recurring across source domains—vertical walls, horizontal floors and ceilings, and slanted outdoor ramps—while the twenty offset exemplars are distributed unevenly across the depth range, with denser coverage below 20 m where indoor annotations are concentrated and sparser coverage for the longer-range outdoor distributions. At test time, the classification logits serve as soft mixture weights that blend exemplar predictions, smoothly interpolating between codebook entries rather than committing to a hard discrete assignment. This soft weighting proves especially beneficial on out-of-distribution evaluation scenes where the true plane parameters may lie between two exemplar centers, allowing the regression residual to compensate for codebook discretization error.

4.4. Geometry-Enhanced Embeddings

To further strengthen the geometric reasoning of plane queries, we introduce auxiliary pixel-level prediction heads that estimate per-pixel depth maps and surface normal maps from the decoder features. The resulting depth and normal embeddings are then fused into the plane queries via cross-attention, allowing each plane query to attend to geometry-rich pixel features. This geometry-enhanced attention mechanism provides the decoder with explicit spatial and orientation cues, improving both segmentation quality and plane parameter accuracy.

4.5. Training Objectives

Our training objective combines multiple loss terms. Following bipartite matching between predicted and ground-truth planes, we apply: (1) a binary cross-entropy loss and a dice loss for plane mask prediction, (2) cross-entropy losses for normal and offset exemplar classification, (3) ℓ_1 regression losses for the normal and offset residuals, and (4) auxiliary losses for the pixel-level depth and surface normal predictions. All loss terms are weighted and summed to form the final training objective.

Training details. We train ZeroPlane with AdamW at a learning rate of 1×10^{-4} and weight decay of 10^{-4}, using a cosine annealing schedule over 50 epochs with a linear warm-up over the first 1,000 iterations. Input images are randomly cropped and resized to 640×480 during training, with horizontal flipping, color jitter, and random grayscale conversion applied as data augmentation to further improve cross-domain robustness. Loss weights are: mask BCE and dice 1.0 each; normal and offset classification cross-entropy 2.0; regression ℓ_1 losses 1.0; auxiliary depth and normal prediction losses 0.1. All models are trained with a batch size of 16 across 8 NVIDIA A100 GPUs. Domain-balanced sam-

pling enforces equal contribution from each of the ten training sources per mini-batch, preventing large-scale datasets such as Taskonomy from dominating the gradient signal.

Inference. At test time, ZeroPlane processes a single RGB image in one forward pass, producing plane instance masks and geometric parameters simultaneously. The DINOv2-B variant runs at approximately 5 frames per second on an NVIDIA A100 GPU at 640×480 resolution, offering a practical trade-off between accuracy and computational cost.

5. Experiments

5.1. Implementation Details

We implement ZeroPlane using three backbone variants: DINOv2 ViT-Base (Ours-DINO-B), DINOv2 ViT-Large (Ours-DINO-L) [11], and DUSt3R (Ours-Dust3R) [23]. All models are trained for 50 epochs with an input resolution of 640×480. For the decoupled geometry prediction, we use $K_n = 7$ normal exemplars and $K_d = 20$ offset exemplars with a split threshold at 20 m. For mixed-dataset training, we combine indoor and outdoor sources from our benchmark to construct a training set of approximately 560K images, as described in Sec. 3. We compare against prior methods including PlaneRCNN [10], PlaneAE [27], PlaneTR [20], and PlaneRecTR [16].

5.2. Zero-Shot Generalization

We evaluate zero-shot generalization on four unseen benchmarks: NYUv2 [18] and 7-Scenes [17] for indoor scenes, and ParallelDomain [7, 12] and ApolloScape [8] for outdoor scenes. Results are reported in Tab. 2.

Under the mixed-training setting (M), our method consistently outperforms PlaneRecTR [16] across nearly all metrics. On NYUv2, Ours-DINO-B achieves a depth recall@0.1 m of 17.86% and a normal recall@10° of 47.58%, compared to 14.29% and 40.96% for PlaneRecTR. The improvements are even more pronounced on outdoor scenes: on ParallelDomain, Ours-DINO-B achieves 44.51% depth recall@3 m and 65.74% normal recall@10°, substantially surpassing PlaneRecTR's 35.49% and 35.11%, respectively. The DINO-L and Dust3R variants offer additional gains, with Ours-Dust3R achieving the best depth recall on NYUv2 (21.20%@0.1 m) and ParallelDomain (46.26%@3 m). On 7-Scenes, the mixed-trained Ours-Dust3R achieves 19.14% depth recall@0.1 m and 36.11% normal recall@10°, surpassing PlaneRecTR by 8.2 and 9.4 percentage points respectively. On ApolloScape, Ours-DINO-L attains the highest overall scores with 46.40% depth recall@3 m and 40.32% normal recall@10°, gains of 3.3 and 7.0 points over the baseline, confirming that larger ViT encoders yield stronger geometric precision on outdoor scenes.

In terms of segmentation quality, mixed-domain training also yields consistent gains in Rand Index (RI) and Segmentation Covering (SC) across all evaluation sets. The improvements are particularly pronounced on outdoor datasets, where the model benefits from exposure to diverse plane scales and spatial distributions during training. On ApolloScape, for instance, the mixed-trained variants produce notably fewer over-segmented regions compared to ScanNet-only baselines, as reflected by lower Variation of Information (VOI) scores. These results underscore that large-scale mixed training does not merely improve geometric accuracy but also strengthens the model's ability to delineate perceptually coherent plane instances. Taken together, the geometry and segmentation metrics confirm that ZeroPlane achieves a well-balanced trade-off between precise plane parameter estimation and high-quality instance-level segmentation across both indoor and outdoor domains.

To better understand the source of gains, we further analyze performance stratified by plane size. Large planes (occupying more than 10% of the image area) benefit most from mixed-domain training, with depth recall improvements of up to 6 percentage points on NYUv2 and 12 points on ParallelDomain. Small planes show more modest gains, consistent with the hypothesis that higher-resolution training and finer-grained feature representations are necessary to fully leverage mixed-domain diversity at fine spatial scales. We also observe that scenes with dominant horizontal planes (floors, road surfaces) show the greatest improvement from outdoor data inclusion, as these surface types are underrepresented in purely indoor training corpora.

Even when trained only on ScanNet (S), our DINO-B model achieves competitive or superior results to PlaneRecTR on NYUv2, with notably better normal recall@10° (39.88% vs. 34.42%), demonstrating that the architectural improvements alone contribute meaningfully.

5.3. Ablation Studies

Component ablation. Tab. 3 presents the contribution of each proposed component, evaluated on NYUv2. Starting from the baseline, adding the DPT pixel decoder yields substantial improvements in both depth and normal recall. The classification-then-regression (Cls-Reg) scheme further improves normal accuracy, while geometry-enhanced attention (Geo-Attn) provides additional gains across all metrics. Under mixed-dataset training, the full model achieves 17.86% depth recall@0.1 m and 47.58% normal recall@10°, compared to 11.65% and 38.15% for the baseline—a relative improvement of over 50% in depth recall and 25% in normal recall.

Dataset combination. Tab. 4 studies the effect of different training data combinations. Training exclusively on indoor data performs well on NYUv2 (18.70% depth recall@0.1 m)

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Table 2. Zero-shot evaluation of different methods or settings on indoor datasets (NYUv2 [18], 7-Scenes [17]) and outdoor datasets (ParallelDomain [7, 12], ApolloScape [8]). (S: trained on ScanNetv1 [5]; S-v2: trained on ScanNetv2, whose training set is much larger than ScanNetv1; M: trained on mixed datasets.)

Evaluation Dataset (Indoor)	Method	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI(↑)	VOI(↓)	SC(↑)	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
NYUv2 [18]	PlaneRCNN (S-v2) [10]	0.84	1.60	0.61	3.04	9.82	**43.41**	9.24	24.12	43.60
	PlaneAE (S) [27]	0.89	1.40	0.68	1.26	3.36	15.06	11.18	26.54	38.52
	PlaneTR (S) [20]	0.89	1.16	0.71	2.14	6.33	28.30	12.42	27.01	34.63
	PlaneRecTR (S) [16]	**0.92**	1.05	**0.75**	3.3	10.13	41.47	16.42	34.42	45.7
	Ours-DINO-B (S)	**0.92**	**1.02**	**0.75**	**4.33**	**11.03**	41.45	**24.05**	**39.88**	**47.35**
	PlaneRecTR (M) [16]	0.91	1.01	0.73	5.77	14.29	52.01	24.97	40.96	54.71
	Ours-DINO-B (M)	**0.92**	**0.93**	**0.75**	8.54	17.86	55.08	37.29	47.58	57.19
	Ours-DINO-L (M)	**0.92**	0.96	**0.75**	9.32	20.29	55.37	38.15	**47.76**	**56.3**
	Ours-Dust3R (M)	**0.92**	0.97	**0.75**	**9.94**	**21.2**	**56.03**	**38.32**	47.6	56.14
7-Scenes [17]	PlaneRecTR (M) [16]	**0.92**	**1.04**	0.78	3.20	10.97	45.30	11.17	26.67	42.17
	Ours-DINO-B (M)	**0.92**	1.06	**0.79**	8.19	17.19	48.13	23.96	35.43	44.17
	Ours-DINO-L (M)	**0.92**	**1.04**	0.78	7.16	16.25	48.74	25.51	**36.32**	**44.49**
	Ours-Dust3R (M)	**0.92**	1.08	0.78	**9.11**	**19.14**	**49.73**	**26.69**	36.11	44.43
Evaluation Dataset (Outdoor)	**Method**	**Plane Segmentation**			**Plane Recall (depth)**			**Plane Recall (normal)**		
		RI(↑)	VOI(↓)	SC(↑)	@1m	@3m	@10m	@5°	@10°	@30°
ParallelDomain [7, 12]	PlaneRecTR (M) [16]	0.92	0.96	0.76	19.11	35.49	51.92	24.17	35.11	53.36
	Ours-DINO-B (M)	0.94	0.65	0.85	25.96	44.51	63.45	53.23	65.74	69.62
	Ours-DINO-L (M)	**0.95**	**0.57**	**0.87**	**27.66**	44.72	**64.0**	52.47	**66.13**	**70.47**
	Ours-Dust3R (M)	0.94	0.69	0.84	27.4	**46.26**	63.19	**53.91**	65.36	69.87
ApolloScape [8]	PlaneRecTR (M) [16]	0.94	0.38	0.91	10.09	43.06	57.06	22.99	33.32	51.88
	Ours-DINO-B (M)	**0.96**	**0.3**	**0.93**	10.28	40.44	57.34	27.56	39.78	56.05
	Ours-DINO-L (M)	**0.96**	**0.3**	**0.93**	**12.26**	**46.4**	**62.08**	**28.51**	**40.32**	**59.87**
	Ours-Dust3R (M)	0.95	0.35	0.92	10.96	44.21	58.9	24.43	37.3	55.15

Table 3. Ablation studies on the contributed components under both single-dataset (ScanNet) training and mixed-dataset training schemes, evaluated on the NYUv2 dataset.

Training datasets	Component			Plane Recall (depth)			Plane Recall (normal)		
	DPT	Cls-Reg	Geo-Attn	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
ScanNet	-	-	-	2.76	8.21	39.37	10.87	29.61	45.1
	✓	-	-	4.17	10.66	41.2	19.78	37.93	47.0
	✓	✓	-	3.88	9.71	41.27	**24.27**	38.73	45.84
	✓	✓	✓	**4.33**	**11.03**	**41.45**	24.05	**39.88**	**47.35**
Mixed datasets	-	-	-	3.9	11.65	48.65	20.54	38.15	53.1
	✓	-	-	6.1	15.65	54.18	29.55	45.23	56.59
	✓	✓	-	7.61	16.93	53.97	36.11	46.36	55.39
	✓	✓	✓	**8.54**	**17.86**	**55.08**	**37.29**	**47.58**	**57.19**

but fails on outdoor scenes (1.19% on ParallelDomain), and vice versa. The full mixed-dataset configuration (560K images) achieves the best balance, reaching 17.86% on NYUv2 and 44.51% on ParallelDomain, demonstrating that large-scale mixed training enables effective cross-domain generalization without sacrificing in-domain performance.

Table 4. Ablation study on training our framework using different combinations of indoor and outdoor datasets.

Evaluation Dataset	Training Datasets	Plane Recall (depth)			Plane Recall (normal)		
		@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2 [18]	Indoor Only	8.41	**18.7**	**55.89**	**37.82**	**47.93**	57.08
	Outdoor Only	0.43	1.71	15.59	7.9	14.68	26.75
	Starter Mixed (110K)	5.24	13.43	46.26	31.2	43.14	51.04
	Full Mixed (560K)	**8.54**	17.86	55.08	37.29	47.58	**57.19**
ParallelDomain [7, 12]	Outdoor Only	**26.72**	43.23	59.87	48.55	59.32	64.43
	Indoor Only	0.81	1.19	3.66	16.94	21.4	37.57
	Starter Mixed (110K)	14.04	35.45	53.53	42.54	55.91	61.36
	Full Mixed (560K)	25.96	**44.51**	**63.45**	**53.23**	**65.74**	**69.62**

Table 5. Comparison on training using low vs. high resolution data, then evaluation on high-resolution groundtruth.

Evaluation Dataset	Training Resolution	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI(↑)	VOI(↓)	SC(↑)	@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2	Low-res	**0.92**	0.93	0.76	**8.35**	**17.78**	55.33	37.41	47.87	57.35
	High-res	**0.92**	**0.84**	**0.77**	7.47	16.19	**57.31**	**40.09**	**50.34**	**59.87**
ParallelDomain	Low-res	**0.94**	0.68	0.84	26.38	44.43	62.68	51.62	64.34	68.43
	High-res	**0.94**	**0.63**	**0.85**	**30.38**	**50.0**	**66.09**	**56.55**	**67.91**	**71.32**

Training resolution. Tab. 5 compares low- and high-resolution training. High-resolution training improves VOI and normal metrics on both NYUv2 (normal recall@10°: 50.34% vs. 47.87%) and ParallelDomain (depth recall@3 m: 50.00% vs. 44.43%), indicating that finer spatial details benefit plane boundary delineation and geometric accuracy, particularly for outdoor scenes.

Backbone comparison. Comparing the three backbone variants in Tab. 2, geometry-aware DUSt3R features yield the highest depth recall on indoor benchmarks (21.20% on NYUv2 and 19.14% on 7-Scenes at @0.1 m), indicating that geometry-pretrained representations provide a useful inductive bias for plane parameter estimation. DINOv2-L achieves the strongest overall performance on outdoor scenes, attaining the best scores on both ParallelDomain and ApolloScape. DINOv2-B already surpasses all prior methods by a large margin at reduced computational cost, confirming that ZeroPlane's core architectural choices—the DPT pixel decoder, decoupled normal/offset prediction, and geometry-enhanced attention—are the primary contributors to improvement rather than backbone scale alone.

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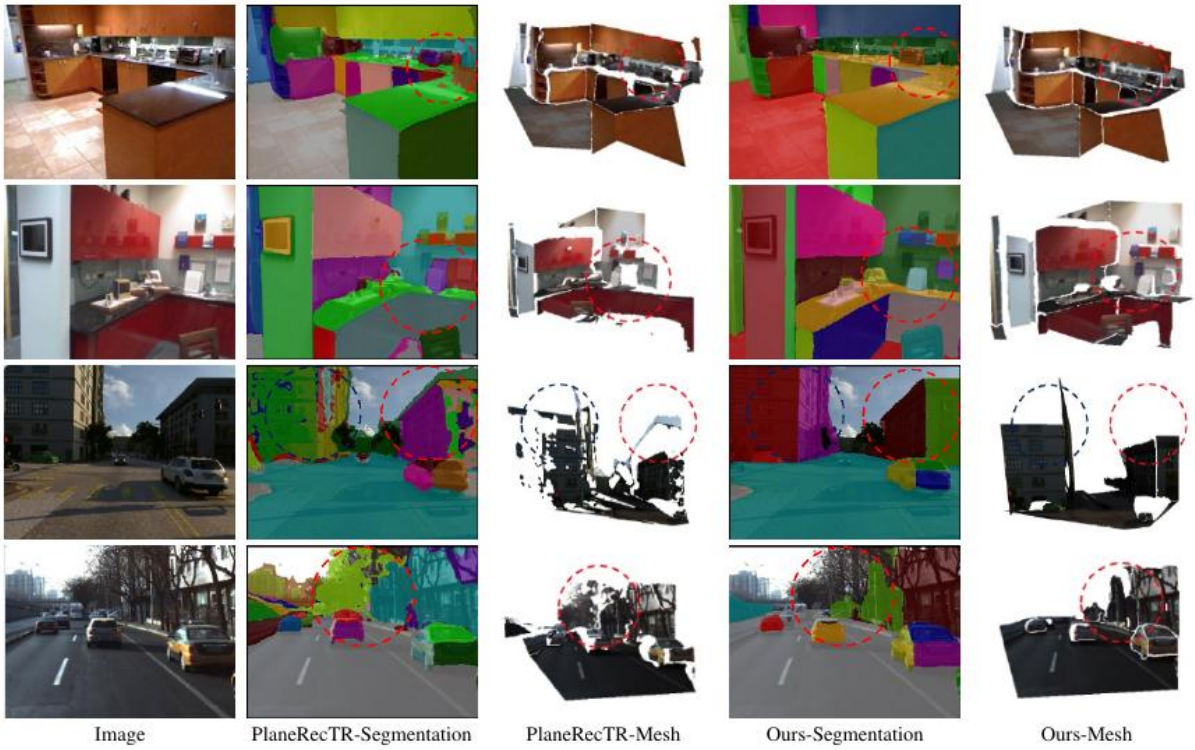

Figure 2. Qualitative results of our mix-trained ZeroPlane model for zero-shot plane segmentation and mesh reconstruction on NYUv2, 7-Scenes, ParallelDomain, and ApolloScape (top to bottom). Our model recovers accurate plane geometry across diverse indoor and outdoor environments.

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5.4. Qualitative Results

Fig. 2 shows qualitative results of our mixed-trained model on all four evaluation datasets. ZeroPlane produces accurate plane segmentations and coherent 3D mesh reconstructions across diverse indoor and outdoor environments, including complex scenes with multiple planes at varying depths and orientations. Compared to the ScanNet-trained baseline, the mixed-trained ZeroPlane yields sharper plane boundaries and more accurate 3D parameters, especially on outdoor scenes where the baseline completely fails to recover meaningful planar structure.

6. Conclusion

We presented ZeroPlane, a single-image 3D plane reconstruction framework designed for in-the-wild generalization across both indoor and outdoor environments. Our key contributions include a DPT-style pixel decoder for multi-scale features, a decoupled exemplar-guided classification-then-regression scheme for robust normal and offset prediction, and geometry-enhanced embeddings that inject depth and normal cues into the plane query decoding process. Alongside the method, we introduced the ZeroPlane Planar Benchmark, a large-scale mixed-domain benchmark spanning 14 datasets with approximately 560K dense plane annotations. Through extensive experiments on four zero-shot benchmarks, we demonstrated that ZeroPlane substantially outperforms prior methods, particularly in geometric accuracy for both plane normals and depth. Our ablation studies confirm the effectiveness of each proposed component and highlight the importance of large-scale mixed-domain training for cross-environment generalization.

Limitations and future work. Despite strong zero-shot results, ZeroPlane has several limitations worth acknowledging. The RANSAC-based annotation pipeline may miss thin or curved surfaces that deviate from the planar assumption, and pseudo-labels generated by Mask2Former introduce label noise for datasets lacking ground-truth semantics. Additionally, the current framework predicts a fixed number of plane queries, which may limit performance on scenes with many small planes. Future work will explore self-supervised annotation refinement, extension to video sequences for temporally consistent reconstruction, and adaptive query mechanisms to better handle scenes with high plane density. The reliance on camera intrinsics at annotation time also means

that evaluation metrics are sensitive to calibration accuracy; future work should explore intrinsic-free formulations that operate robustly under approximate or unknown camera models.

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